


Fig. 8.9 Limit of a function

When compared to the case of a function of single variable, for a function of two variables, there is a subtle depth in the limiting process. Here the values of $F(x,y)$ should approach the same value $L$, as $(x,y)$ approaches $(u,v)$ along **every possible path to** $(u,v)$ (including paths that are not straight lines). Fig.8.9 explains the limiting process.

**All the rules for limits (limit theorems) for functions of one variable also hold true for functions of several variables.**

Now, following the idea of continuity for functions of one variable, we define continuity of a function of two variables.

**Definition 8.7 (Continuity)**

Suppose that $A=\{(x,y)|a<x<b, c<y<d\}\subset\mathbb{R}^2, F:A\to\mathbb{R}$. We say that $F$ is continuous at $(u,v)$ if the following hold :

(1) $F$ is defined at $(u,v)$ (2) $\lim\limits_{(x,y)\to(u,v)} F(x,y)=L$ exists (3) $L=F(u,v)$.

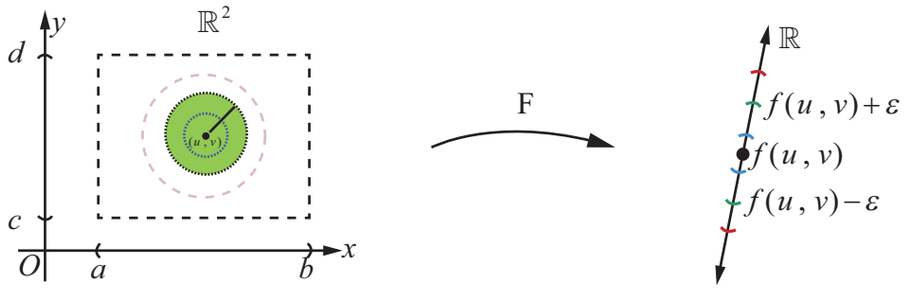



Fig. 8.10 Continuity of a function

**Remark**

(1) In Fig. 8.10 taking $L=F(u,v)$ will illustrate continuity at $(u,v)$.

(2) Continuity for $f(x_1,x_2,\ldots,x_n)$ is also defined similarly as defined above.

Let us consider few examples as illustrations to understand continuity of functions of two variables.

**Example 8.8**

Let $f(x,y)=\dfrac{3x-5y+8}{x^2+y^2+1}$ for all $(x,y)\in\mathbb{R}^2$. Show that $f$ is continuous on $\mathbb{R}^2$.



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